UBER DELHI-NCR 2024

Individual Case Analysis

From raw booking events to marketplace diagnosis, competitive vulnerability, and management action

What we are building

A 30-mark individual case, not a group ML project.

Central management question

Where is Uber failing to convert demand into completed rides — and where should management intervene first?

- Use 2024 data as the only marketplace evidence
- Identify internal vulnerability, not competitor market share
- Reward diagnosis + prioritisation + defensible action
- Keep ML as an optional extension, not the core

Data — **Metric** — **Pattern** — **Diagnosis** — **Action**

Why 2024 matters

Treat the dataset as a historical baseline, not a live 2026 market-share file.

Observation	What the Uber data directly shows in 2024
Inference	What pattern probably means operationally
Hypothesis	What might explain the pattern
Causal claim	Requires additional evidence or experiment

Rule for students: do not claim competitor behaviour without competitor data.

The dataset: what it can and cannot see

Each row is a booking event, not a complete view of the driver marketplace.

Fields available

- Date / Time / Booking ID / Customer ID
- Booking Status
- Vehicle Type
- Pickup / Drop Location
- Avg VTAT / Avg CTAT
- Cancellation and incomplete reasons
- Booking Value / Distance / Ratings / Payment

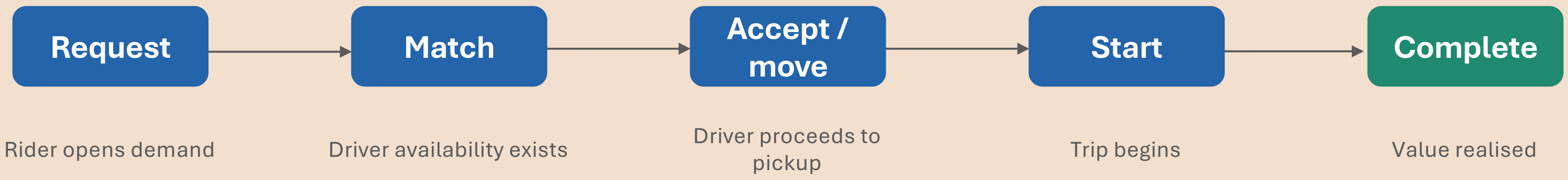
Missing for causality

- Driver ID and online supply
- Acceptance and rejection attempts
- Driver earnings, commission, incentives
- Surge and quoted ETA
- GPS movement and dead kilometres
- Competitor supply or price

Therefore: we can diagnose vulnerability, but we cannot prove every cause.

Marketplace funnel

The business problem is conversion of demand into successful rides.



Available status proxies

Completed | No Driver Found | Cancelled | Incomplete

Core KPIs students must define

Denominator discipline is the first analytical test.

Completion Rate

Completed / Total Booking Requests

No Driver Rate

No Driver Found / Total Booking Requests

Driver Cancel Rate

Cancelled by Driver / Total Booking Requests

Customer Cancel Rate

Cancelled by Customer / Total Booking Requests

Incomplete Rate

Incomplete / Total Booking Requests

Wait Pain

Avg VTAT + / or Avg CTAT by segment

Never treat null fare / rating / distance on failed rides as zero unless the metric explicitly requires it.

Micro-market lens

Annual Delhi-NCR averages hide the real operational problem.

Bad grain

Delhi-NCR overall completion rate

Useful grain

Pickup Location
× Time Bucket
× Vehicle Type

Even better

Origin → Destination
× Hour
× Product

- Micro-markets make action possible: supply, pricing, pickup-zone ops, cancellation controls.
- Example framing: Auto + Gurgaon + evening is more actionable than “NCR cancellations are high.”

Vehicle-type strategy

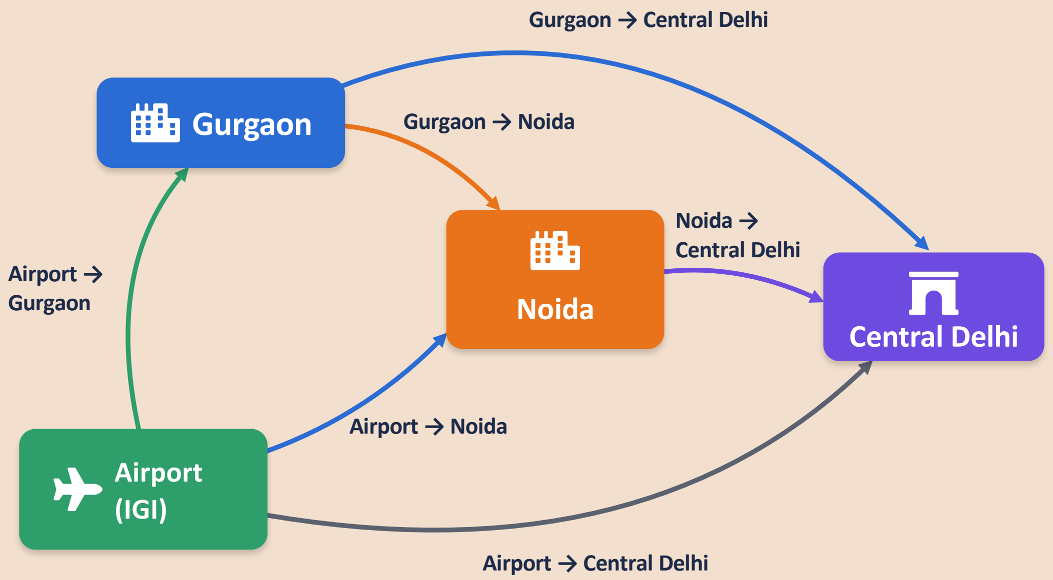
Each product category has a different competitive vulnerability.

Vehicle	Why it matters	Key vulnerability test
Bike / eBike	Fast, low-ticket trips	Speed + supply density
Auto	High urban demand	Availability + cancellation friction
Go Mini / Sedan	Core cab marketplace	Price, ETA, driver economics
Premier / XL	Premium use cases	Service reliability + comfort

Product averages should not be mixed blindly: Auto failure and Premier failure imply different interventions.

Origin – Destination (OD) View

Pickup and drop locations turn the dataset into a mobility network, helping us identify key corridors.



Demand volume alone is not enough

A location may have high bookings but low completion rates.

A corridor can be high-value and operationally broken

Some OD pairs have strong demand but high cancellations or long waiting times.

Prioritise high demand + high failure + high waiting pain

These corridors represent the biggest opportunity for improvement (or for a competitor to enter).

What is an OD pair?

Each booking has a pickup location (origin) and a drop location (destination). By analysing origin–destination corridors, we can understand how trips between specific areas perform, rather than just looking at individual locations.

Example Corridor Analysis

Illustrative numbers (not from the actual dataset)

Origin → Destination	Bookings	Completion Rate	No Driver Found	Cancellation Rate	Avg CTAT (min)	₹/km
Gurgaon → Noida	8,000	68%	14%	18%	31	₹18
Gurgaon → Central Delhi	12,000	91%	2%	7%	17	₹21
Airport → Gurgaon	6,000	95%	1%	4%	12	₹28
Airport → Noida	3,500	72%	10%	15%	26	₹19
Noida → Central Delhi	7,200	85%	5%	10%	22	₹20

Gurgaon → Noida shows high demand but low completion and high waiting time, making it a vulnerable corridor.

The hidden cancellation trap

Reported status may not equal the true behavioural cause.

Booking Status
Cancelled by Customer
→
Reason
“Driver asked to cancel”
→
Business truth?
Driver-induced failure

Better metric

Effective Driver-Induced Failure = Official Driver Cancellation + Customer Cancellation where reason suggests driver-induced behaviour

Competitive vulnerability score

A student may build a transparent prioritisation framework.

Possible formulation

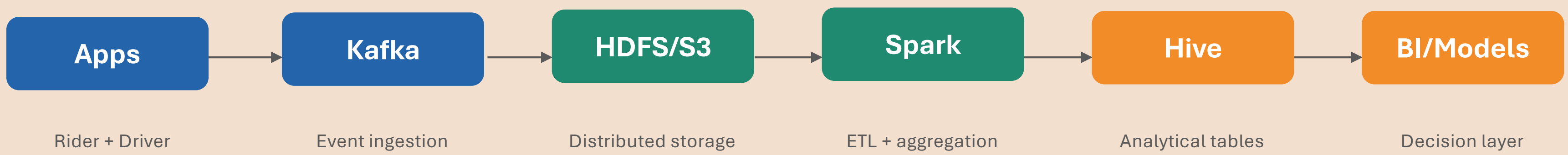
Vulnerability = Demand × Failure Rate × Waiting Pain × Economic Attractiveness

Demand	Failure	Pain	Value
Bookings / request density	No driver + cancellations + incomplete	VTAT / CTAT / rating signal	Booking value / distance / ₹ per km

Do not make this formula mandatory. Reward transparency, normalization, weights, and business defence.

The data architecture question

Keep Big Data technology connected to the business case.



- Assume the CSV is a sample of hundreds of millions of monthly marketplace events
- Ask: partitioning, batch vs streaming, fault tolerance, and reproducibility
- Do not ask students to define Hadoop by memory; make them design how this analysis scales

Data quality: nulls are not all equal

Students must distinguish structural nulls from broken data.

Structural null
Expected because of status

Missing / invalid
Contradicts business logic

Ambiguous
Needs rule + explanation

Example validation rules

- If Booking Status = Completed, distance and booking value should normally exist
- If No Driver Found, fare, distance and ratings may logically be null
- Ratings should be within valid range
- Booking IDs should not duplicate without explanation
- Cancellation reason should align with cancellation status

Appendix

30 marks, three questions

The case should be challenging but feasible as group work.

PySpark / Databricks blueprint

Illustrative structure; not a syntax-memory test.

```
df = spark.read.csv(path, header=True, inferSchema=True)

df2 = (df
 .withColumn("ts", to_timestamp(concat_ws(" ", "Date", "Time")))
 .withColumn("hour", hour("ts"))
 .withColumn("dow", date_format("ts", "E"))
 .withColumn("time_bucket", when(col("hour").between(7,10), "AM Peak")
   .when(col("hour").between(17,21), "PM Peak").otherwise("Other"))
 .withColumn("completed", (col("Booking Status") == "Completed").cast("int"))
 .withColumn("no_driver", (col("Booking Status") == "No Driver Found").cast("int"))
 .withColumn("driver_cancel", (col("Booking Status") == "Cancelled by Driver").cast("int")))

seg = df2.groupBy("Pickup Location","Vehicle Type","time_bucket")
 .agg(count("*").alias("requests"), avg("completed").alias("completion_rate"),
      avg("no_driver").alias("no_driver_rate"), avg("driver_cancel").alias("driver_cancel_rate"),
      avg("Avg CTAT").alias("avg_ctat"))
```

How to use a large LLM here

Use it as an assessment co-designer — with strong guardrails.

- Tell the model it is designing an individual 30-mark case, not a generic analytics project
- Force source-of-truth rules: 2024 dataset only, no unsupported competitor claims
- Require Parts A–G: case sheet, solution guide, rubric, blueprint, code, red flags, bonus
- Ask it to audit: exact marks, CLO coverage, feasibility, no hidden assumptions

Large model ≠ correct model. Guardrails produce academic discipline.

Best use: generate a first faculty draft, then you validate it against the dataset and rubric.

The final standard

A strong student answer identifies one micro-market to fix tomorrow.

Where | When | Vehicle | Problem | Evidence | Action | Unknown